

Detecting Face-Swap Deepfakes by Verifying Skeletal Gait Dynamics

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Abstract—Face-swap generators synthesise a face and composite it back into an otherwise unmodified frame. The body they leave behind still walks the way the source person walks. We verify the claimed identity against the one signal the generator never touches. Our system extracts a 78-dimensional skeletal gait descriptor per frame—hip-centred coordinates, six joint flexion angles and velocities over 12 biomechanically selected MediaPipe landmarks—and classifies the per-timestep discrepancy against the claimed identity’s enrolled signature with a temporal convolutional verification network. On GaitDeepfake-13, a 13-subject corpus of 66 walking videos expanded to 1,056 by augmentation, under strict leave-one-subject-out cross-validation, it attains a pooled ROC-AUC of 94.95% over 2,240 verification pairs (per-fold $95.10\% \pm 3.08\%$), accuracy $87.04\% \pm 3.65\%$ and an equal error rate of $12.27\% \pm 3.80\%$. Gradient attribution rests the decision on shoulder counterrotation and distal heel-strike kinematics, with joint angles carrying $1.94\times$ their proportional share despite occupying 6 of 78 dimensions. On three FaceFusion-generated face-swaps the system rejected the claimed identity in all three cases and matched the true body source above 0.9998 similarity—an existence check, not a powered claim. An ablation over 13 folds and 3 seeds finds that adding a CNN, BiLSTM or Transformer encoder to the decision path costs 2.5 to 5.9 ROC-AUC points (all $p < 10^{-3}$, paired), so the network’s simplicity is a result rather than a concession.

Index Terms—Media forensics, multimedia security, deepfake detection, gait analysis, behavioural biometrics, identity verification, pose estimation, explainable AI.

I. INTRODUCTION

The dominant paradigm in video deepfake detection looks for evidence of manipulation inside the manipulated region: detectors learn blending boundaries, frequency-domain irregularities, warping residues and temporal flicker in and around the face crop [1]–[5]. The paradigm is productive and structurally fragile at once, because it places the detector in direct opposition to the generator’s objective. Every improvement in synthesis quality erodes precisely the signal the detector depends on, which is why cross-dataset generalisation remains the field’s most stubborn problem—methods approaching saturation within a training corpus lose substantial ground on unseen manipulations and generators [1], [3], [6].

We take the opposite stance: instead of interrogating the region the adversary controls, we interrogate a region the adversary does not modify at all. The observation is mechanical rather than statistical. Contemporary face-swap pipelines—FaceFusion with InSwapper [7], [8], SimSwap [9] and their descendants—detect a face, synthesise a replacement, and

composite it back through a mask confined to the facial region, carrying the torso, limbs and background through from the source footage modulo recompression. A face-swapped video of person B built on footage of person A therefore contains B ’s face and A ’s *gait*. Human locomotion is constrained by skeletal proportion, mass distribution and individual motor control, and is a well-established biometric in its own right [10]–[14]: the forgery carries, in plain sight, an identity signal contradicting its own claim.

A. Problem Formulation

We treat detection as identity verification rather than binary artefact classification. Given a video V and a claimed identity I_c drawn from an enrolled gallery, the system computes

$$f(V, I_c) \rightarrow \begin{cases} \text{AUTHENTIC} & \text{if the gait in } V \text{ matches } I_c, \\ \text{DEEPFAKE} & \text{otherwise.} \end{cases} \quad (1)$$

Consequently the system detects an *identity inconsistency*, not the presence of synthesis. It requires an enrolled reference and will not flag a wholly synthetic person with no claimed identity. Section VII returns to what this rules in and out.

B. Contributions

- 1) A **gait-based verification formulation** of face-swap detection whose discriminative signal lies outside the generator’s output support (Section III).
- 2) A **difference-based temporal verification network** that compares sequences before pooling, avoiding the representation collapse a Siamese formulation suffers at this corpus size (Section III-E).
- 3) A **subject-disjoint evaluation** over 13 leave-one-subject-out folds and 2,240 verification pairs, with per-fold dispersion and two operating points (Sections IV–V).
- 4) **Gradient attribution over the skeleton**, quantifying the disproportionate value of the six joint-angle dimensions (Section VI).
- 5) **An ablation that refutes the alternative**: adding a CNN, BiLSTM or Transformer stack costs 2.5 to 5.9 ROC-AUC points (all $p < 10^{-3}$, paired), and removing the raw comparison collapses performance to chance (Section V-C).

II. RELATED WORK

A. Face-Based Deepfake Detection

Detection of facial manipulation has converged on spatial artefact modelling, temporal coherence modelling, or a hybrid. Multi-attentional detection [2] treats forgery as fine-grained classification. RECCE [1] uses reconstruction error of genuine faces as the forgery cue, observing that earlier methods degrade towards approximately 60% AUC on unseen data and reporting roughly 70.9% cross-dataset AUC on Celeb-DF-v2. FTCN [3] removes spatial convolution capacity to force reliance on temporal incoherence, reaching roughly 86–89% AUC on Celeb-DF cross-dataset, and AltFreezing [4] alternately freezes spatial and temporal weights in a 3D ConvNet for 89.5% on the same setting; FSBI [5] synthesises training forgeries via frequency-enhanced self-blending. In-corpus performance is near saturated while cross-corpus performance is not, and the gap is generator-dependent. Our approach is complementary rather than competitive: it never examines the facial region, so its failure modes are uncorrelated with every method here.

B. Skeleton-Based Gait Recognition

Model-based gait recognition operates on pose sequences rather than silhouettes, conferring robustness to clothing and carried objects. GaitGraph [10] applies graph convolution to skeleton sequences; GaitPT [11] reaches 82.6% average accuracy on CASIA-B from skeletons alone; GaitFormer [12] reports 92.5% on CASIA-B after multi-task pretraining. Skeletal kinematics thus carry sufficient identity information for recognition at scale; we redirect that premise to adjudicating a specific claimed identity, a verification problem with its own metric conventions [15].

C. Behavioural and Whole-Body Forgery Detection

Behavioural biometrics entered deepfake detection with Agarwal *et al.* [16], who modelled correlations among facial action units and head motion, later extending beyond the face to gestural and vocal mannerisms [17]. Attention has since moved to the whole body. Human Action CLIPs [18] detects AI-generated human motion; TT-DF [19] benchmarks human body forgery detection from pose and motion inconsistency; Chapariniya *et al.* [20] identify people from whole-body conversational keypoints, framed against face-reenactment threats. Closest to this work, DeAndres-Tame *et al.* [21] evaluate whether generative human-animation models preserve gait identity cues and find that they do not—an evaluation study rather than a deployed detector, but one that independently supports our premise.

D. Positioning

Under-explored is *locomotion gait-cycle skeletal dynamics as the primary discriminative feature* for verifying a claimed identity in a face-swapped video under a subject-disjoint protocol. The nearest work [21] measures gait fidelity in generated animation without building a detector; the nearest

detectors [18], [19] target whole-body synthesis, not face-swap, where the body is authentic and only the claim is false.

III. PROPOSED METHOD

A. Threat Model

We assume an adversary who produces a face-swapped video: a genuine recording of a source subject with a target subject’s face composited into the facial region, presented under the target’s identity. The adversary may use any face-swap model and any face-enhancement post-process, and may recompress the result, but does *not* re-synthesise or retime body motion.

Under this model the detector’s task reduces to Eq. 1, and its robustness has an unusual character: it is invariant to the quality of the face swap, because it never observes the swapped region (Fig. 2). Section VII-C discusses adversaries violating the last assumption.

B. Dataset

Evaluation uses GaitDeepfake-13 [22], collected for this work. Thirteen subjects were recorded walking at natural pace in controlled indoor conditions, from frontal and side viewpoints, yielding 66 original recordings (3–5 per subject) at 30 fps and 720p–1080p. Subjects were students aged approximately 19–25 of both sexes. Augmentation expands the corpus to 1,056 sequences; each is resampled to $T = 60$ frames (≈ 2 s) of 78-dimensional descriptors. The corpus also contains 3 face-swapped test clips (FaceFusion / InSwapper), and the leave-one-subject-out protocol yields 2,240 verification pairs, of which 1,120 are genuine.

Each recording was expanded $16\times$ ($66\times 16 = 1,056$) by transformations that leave the skeletal trajectory unchanged: photometric (blur, brightness ± 0.2 – 0.3 , contrast, colour jitter, grayscale, noise), anatomically plausible geometric (flip, rotation $\leq 10^\circ$, zoom), and temporal (speed $0.8\times/1.2\times$, reversal, exploiting the approximate time-reversal symmetry of the gait cycle [13]), plus a combined pass.

C. Pose Estimation and Feature Extraction

Pose is estimated with MediaPipe Pose Landmarker [23] (pose_landmarker_lite, FP16, per-frame IMAGE mode, detection and tracking confidence 0.5), producing 33 three-dimensional landmarks per frame. We retain 12: the bilateral shoulders, hips, knees, ankles, heels and foot indices. The selection is driven by gait biomechanics [13], [14]—the lower-limb chain from hip through knee, ankle, heel and foot index spans the events that define the gait cycle (heel strike, stance, toe-off), while the shoulders capture the trunk counter-motion that balances angular momentum during swing. Facial, elbow and wrist landmarks are discarded; the facial ones on principle, since they lie inside the region the adversary controls.

Video pose agrees with Vicon motion capture to within a few degrees (errors 4.0° – 7.4° at hip, knee, ankle [24]), though that study used OpenPose, not MediaPipe.

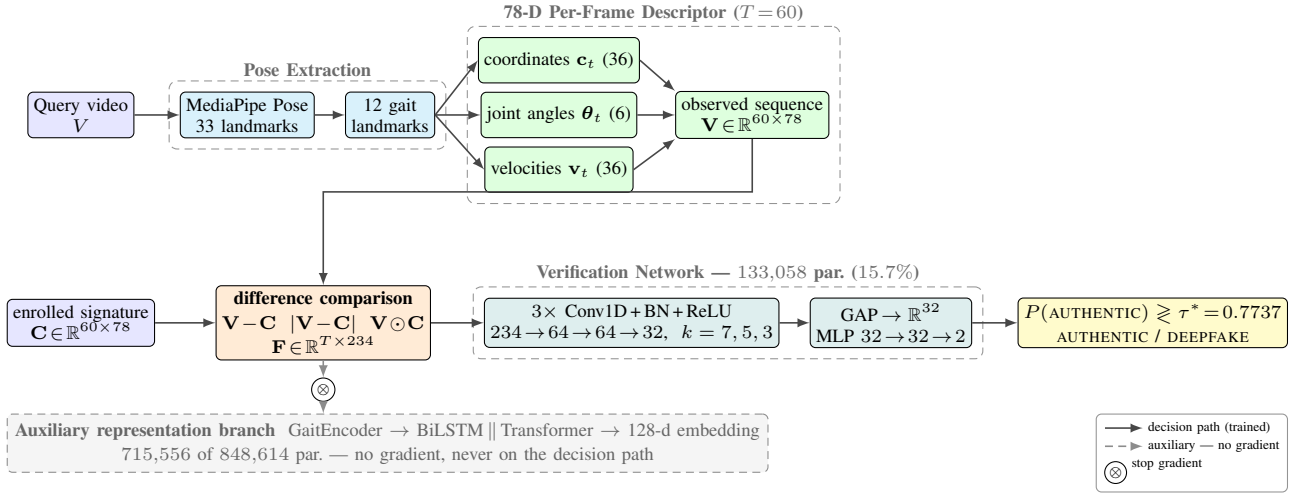


Fig. 1. End-to-end verification pipeline. The observed sequence is compared against the claimed identity’s enrolled signature *before* pooling, and a three-stage temporal CNN classifies the $T \times 234$ comparison tensor. The auxiliary branch holds 715,556 of the model’s 848,614 parameters but receives no gradient and never reaches the decision path (Section III-E2).

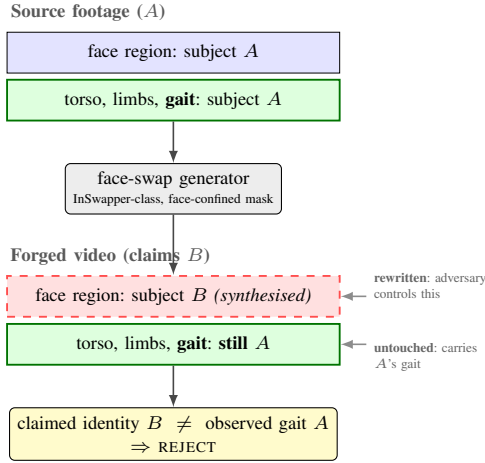


Fig. 2. The asymmetry the method exploits. The generator rewrites only the face-confined band, so a forgery claiming subject B still walks with subject A ’s gait. The generator never models locomotion, so this inconsistency is structural, not an artefact better synthesis will remove.

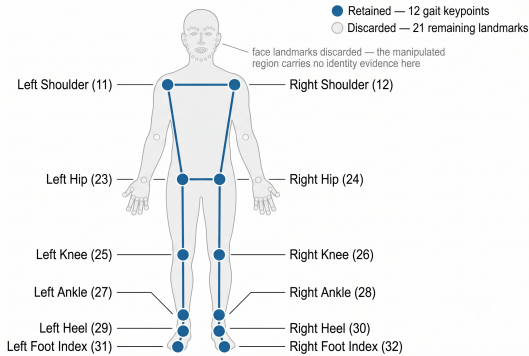


Fig. 3. The 12 retained gait keypoints (blue) of the 33 MediaPipe landmarks; facial, elbow and wrist landmarks are discarded.

For frame t the descriptor concatenates three blocks, $\mathbf{f}_t = [\mathbf{c}_t \parallel \boldsymbol{\theta}_t \parallel \mathbf{v}_t] \in \mathbb{R}^{78}$. *Normalised coordinates* $\mathbf{c}_t \in \mathbb{R}^{36}$ are the 12 landmark positions re-expressed relative to the pelvis

centre, $\hat{\mathbf{p}}_i^{(t)} = \mathbf{p}_i^{(t)} - \frac{1}{2} (\mathbf{p}_{L_Hip}^{(t)} + \mathbf{p}_{R_Hip}^{(t)})$, $i = 1, \dots, 12$, removing translation and much of the scale variation induced by camera distance [10], [11]. *Joint angles* $\boldsymbol{\theta}_t \in \mathbb{R}^6$ are three-point flexion angles

$$\theta_{abc} = \arccos \left(\frac{(\mathbf{p}_a - \mathbf{p}_b) \cdot (\mathbf{p}_c - \mathbf{p}_b)}{\|\mathbf{p}_a - \mathbf{p}_b\| \|\mathbf{p}_c - \mathbf{p}_b\|} \right), \quad (2)$$

evaluated as $\theta(\text{hip, knee, ankle})$, $\theta(\text{shoulder, hip, knee})$ and $\theta(\text{knee, ankle, foot})$ bilaterally; they are scale-free by construction and encode biomechanical constraint directly. *Velocities* $\mathbf{v}_t = \mathbf{c}_t - \mathbf{c}_{t-1} \in \mathbb{R}^{36}$ are first-order differences with $\mathbf{v}_1 = \mathbf{0}$, capturing the rate structure of the gait cycle.

Sequences are resampled to $T = 60$ frames—longer uniformly subsampled, shorter zero-padded, clips with fewer than 10 valid frames rejected—spanning roughly two seconds at 30 fps, or one to two complete gait cycles [13]. Each dimension is then z-score normalised, $\hat{f}_j = (f_j - \mu_j) / \max(\sigma_j, 10^{-6})$, since the blocks occupy incommensurate scales.

D. Identity Enrolment

Each identity I is enrolled by averaging the descriptors of its original (non-augmented) recordings per timestep,

$$\bar{\mathbf{f}}_t^{(I)} = \frac{1}{|V_I|} \sum_{v \in V_I} \mathbf{f}_t^{(v)}, \quad t = 1, \dots, 60, \quad (3)$$

yielding a 60×78 signature per subject. Averaging suppresses per-take noise while preserving the subject’s characteristic temporal profile. The signature is used as a full sequence, not collapsed to a vector—the verification network consumes it as a second input stream.

E. Verification Network

The network classifies the discrepancy between the observed sequence $\mathbf{V} \in \mathbb{R}^{T \times 78}$ and the claimed identity’s signature $\mathbf{C} \in \mathbb{R}^{T \times 78}$ directly, without first reducing either to a

pooled embedding. Three per-timestep comparison signals are concatenated,

$$\mathbf{F} = [(\mathbf{V} - \mathbf{C}) \parallel |\mathbf{V} - \mathbf{C}| \parallel (\mathbf{V} \odot \mathbf{C})] \in \mathbb{R}^{T \times 234}, \quad (4)$$

where the signed difference localises *where* and *in which direction* the sequences diverge, the absolute difference supplies magnitude invariant to sign, and the elementwise product responds where the two agree. $\mathbf{F}$ is transposed to channel-first order and passes through three 1D convolutional stages with decreasing receptive fields ($234 \rightarrow 64$, $k = 7$; $64 \rightarrow 64$, $k = 5$; $64 \rightarrow 32$, $k = 3$), each with batch normalisation and ReLU, and dropout on the first two. Global average pooling over time yields a 32-dimensional summary, which a two-layer MLP ($32 \rightarrow 32 \rightarrow 2$) maps to two logits. The verdict score is $P(\text{AUTHENTIC}) = \text{softmax}(\cdot)_1$.

1) *Why differences rather than embedding distances*: Our initial design followed the standard Siamese template: encode both sequences with shared weights, pool to embeddings, score their distance. On this corpus it collapsed—the encoder converged to a near-constant embedding, driving all pairwise distances towards a single value and the classifier towards the majority prediction. With 13 identities and a per-timestep signature that is itself an average of few recordings, pooling discards precisely the fine temporal structure separating individuals. Comparing sequences before pooling avoids this: discriminative information is preserved at full temporal resolution, and the pooling that remains happens after the comparison, on evidence rather than identity. Section V-C tests this directly.

2) *Auxiliary representation branch*: The trained checkpoint also contains a representation branch retained from the Siamese design and used for enrolment analysis and diagnostics: a residual 1D CNN encoder ($78 \rightarrow 64 \rightarrow 128$) feeding a dual-path temporal model whose BiLSTM ($h = 64$, bidirectional) and Transformer encoder ($d_{\text{model}} = 128$, 4 heads, 2 layers) are fused into a 128-dimensional embedding.

This affects how Section V should be read. Under the cross-entropy objective applied to the verification logits, gradients reach only the pairwise-comparison path; the representation branch receives no learning signal from the training objective. Of the model’s 848,614 parameters, 133,058 (15.7%) lie on the decision path—131,936 in the temporal CNN and 1,122 in the classifier MLP. All quantitative results below are therefore properties of the difference-based verification network, not of the CNN–BiLSTM–Transformer stack. Section V-C asks whether that stack *should* be connected, and finds that it should not.

F. Training

Training pairs are sampled with a 50/50 balance: half pair a video with its own identity’s signature (label AUTHENTIC), half with a uniformly sampled different identity’s signature (label DEEFAKE). This simulates the identity-mismatch condition a face swap induces and prevents the degenerate constant-prediction solution. Optimisation uses AdamW [25] at 10^{-3} with weight decay 10^{-4} , batch size 16, gradient-norm

clipping at 1.0, dropout 0.1, and ReduceLROnPlateau (factor 0.5, patience 7). Early stopping monitors validation accuracy with patience 20 and $\delta_{\min} = 10^{-3}$. The loss is class-weighted cross-entropy over a maximum of 50 epochs (30 under LOOCV), on an NVIDIA RTX 3050 (6 GB, CUDA 12.4). The single-split reference model reached 93.92% validation accuracy at epoch 46 against 98.38% training accuracy.

IV. EXPERIMENTAL SETUP

A. Leave-One-Subject-Out Cross-Validation

The primary protocol is 13-fold leave-one-subject-out cross-validation. In fold k , every recording of subject s_k —original and all 15 augmentations—is withheld, normalisation statistics are recomputed from the remaining 12 subjects, those 12 identities are re-enrolled from the training split alone, and a model is trained from scratch for 30 epochs before being evaluated on the held-out subject’s balanced genuine/impostor pairs.

Subject-level rather than video-level partitioning is a requirement, not a refinement: because augmented clips derive from originals, a video-level split would place 15 near-duplicates of a test clip into training and yield an estimate with no bearing on unseen subjects. LOOCV suits small-cohort biometric evaluation [15], with the caveat that its variance is high at this cohort size [26]. The full 13-fold procedure took 1,228 s.

1) *A protocol detail, stated up front*: The evaluation script producing our headline LOOCV figures normalises the held-out subject using that subject’s own statistics rather than the training split’s—a transductive choice, not the one a strict reading of the protocol implies. We measured what it is worth by running the deployed model both ways across all 13 folds: training statistics yield $94.67\% \pm 3.06\%$ ROC-AUC against $94.97\% \pm 2.90\%$ for the shipped behaviour, a difference of 0.30 points that is not significant ($p = 0.73$, paired, $n = 13$). The headline result is therefore insensitive to this choice, and all ablation comparisons use training statistics throughout.

B. Metrics and Operating Points

We report ROC-AUC as a threshold-free measure of separability [27]; equal error rate, the biometric convention [15]; and accuracy, precision, recall and F1 at an explicit operating point.

Two aggregation modes are reported. The *per-fold* statistic is the mean and standard deviation of a metric computed within each of the 13 folds (variation by subject); the *pooled* statistic is computed once over all 2,240 scores (behaviour on the population). They differ—pooled AUC is 94.95% against a per-fold mean of 95.10%—because folds carry unequal sample counts and pooling forces one global ranking across models trained on different splits. Two operating points are likewise reported: the native $\tau = 0.5$ argmax decision, at which the per-fold metrics of Table I were computed, and the deployed threshold chosen by Youden’s J [28], $\tau^* = \arg \max_{\tau} [\text{TPR}(\tau) - \text{FPR}(\tau)]$, on the pooled LOOCV ROC,

TABLE I
PER-FOLD LOOCV RESULTS OVER 13 SUBJECT-HELD-OUT FOLDS AND
2,240 VERIFICATION PAIRS (HELD-OUT SUBJECT; %)

Subj.	Acc	F1	Prec	Rec	AUC	EER	n
S1	89.29	90.00	84.38	96.43	98.16	6.55	168
S2	87.65	87.72	87.21	88.24	95.04	12.94	170
S3	88.24	88.10	89.16	87.06	95.54	11.76	170
S4	82.14	81.01	86.49	76.19	90.87	19.05	168
S5	84.80	84.26	87.37	81.37	92.59	15.69	204
S6	86.47	84.97	95.59	76.47	96.71	9.41	170
S7	88.24	88.76	84.95	92.94	95.70	9.41	170
S8	88.24	88.64	85.71	91.76	96.64	12.94	170
S9	86.47	87.57	81.00	95.29	95.71	11.76	170
S10	90.00	89.94	90.48	89.41	97.20	9.41	170
S11	90.00	90.50	86.17	95.29	96.62	11.76	170
S12	77.65	78.65	75.27	82.35	87.00	20.00	170
S13	92.35	92.49	90.91	94.12	98.53	8.82	170
Mean	87.04	87.12	86.51	88.23	95.10	12.27	
Std	3.65	3.77	4.72	6.82	3.08	3.80	
Pooled	87.01	87.15	86.20	88.12	94.95	12.77	2240

Youden-optimal operating point on the pooled ROC: $\tau^* = 0.7737$,
TPR = 83.57%, FPR = 8.48%, $J = 0.7509$.

giving $\tau^* = 0.7737$ with TPR = 83.57%, FPR = 8.48% and $J = 0.7509$.

C. Ablation and Face-Swap Protocols

Five variants were trained under the full LOOCV protocol at three random seeds, differing only in how the observed and claimed sequences are encoded before comparison. All share folds, seeds, schedule and downstream classifier, so each compares to the deployed model pairwise over $13 \times 3 = 39$ matched observations.

Face-swapped clips were generated with FaceFusion [7] using the InSwapper inswapper_128_fp16 model [8] under a face-only mask preserving body, neck and shoulders. Each composites one enrolled subject’s face onto another’s walking footage; presented with the face subject as claimed identity, a correct system should reject, because the locomotion belongs to the body subject. Gait preservation through the swap is assessed by comparing MediaPipe features before and after manipulation under three criteria—PCK@0.05 $\geq 90\%$, mean cosine similarity of hip-normalised pose vectors ≥ 0.95 , and Pearson correlation of step-width and symmetry time series ≥ 0.85 —by majority vote.

V. RESULTS

A. Aggregate and Per-Fold Performance

Table I reports the 13-fold result. The system separates genuine from impostor pairs with a pooled ROC-AUC of 94.95% and, at the native operating point, classifies 87.04% $\pm$ 3.65% of held-out pairs correctly with an F1 of 87.12% $\pm$ 3.77%; the equal error rate is 12.27% $\pm$ 3.80% per fold and 12.77% pooled. Recall carries the largest dispersion (± 6.82), roughly twice that of AUC: on subject-held-out folds, how readily an individual’s genuine pairs are accepted depends more on who that individual is than overall separability does.

The fold-level detail is what an aggregate $\pm \sigma$ suppresses. The distribution is unimodal with a single clear low outlier: Subject 12 yields 87.00% AUC and 77.65% accuracy, roughly

2.6 standard deviations below the mean, while the remaining twelve folds fall between 82.14% and 92.35%. Removing that fold would raise mean accuracy to 87.83%; we report it included. Fold ordering is stable across metrics, pointing to a property of those subjects’ gait rather than threshold artefacts, and precision and recall trade off in opposite directions across folds (Subject 6: 95.59% precision, 76.47% recall; Subject 9: 81.00% precision, 95.29% recall), so no single global threshold suits every subject.

B. Operating Points and Error Structure

Fig. 4 (left) shows the pooled ROC; per-fold curves range from Subject 13 (0.985) to Subject 12 (0.870), with an envelope that is narrow mid-range but widens near the origin, so low-false-accept operation is where subject identity matters most. At the native $\tau = 0.5$ the system produces 987 true positives, 962 true negatives, 158 false positives and 133 false negatives (87.01% accuracy). The Youden-optimal $\tau^* = 0.7737$ moves 63 pairs from false-positive to true-negative at the cost of 51 additional false negatives: false positives fall from 158 to 95, precision rises from 86.20% to 90.79% and recall falls from 88.12% to 83.57%. Which is preferable is a policy question; in forensic screening a false positive means accepting a forgery, so the stricter threshold is the default. The 987/962/158/133 matrix is the $\tau = 0.5$ matrix and should not be quoted as the Youden one. Both classes are strongly saturated—genuine median score 0.999, impostor 0.0007—so only 86 pairs (3.8%) lie between the equal-error and Youden thresholds: threshold choice redistributes a small, hard residue, and reported confidence values should not be read as calibrated probabilities.

C. Architectural Ablation

The dual-path stack of Section III-E2 is inert in the deployed checkpoint; the obvious question is whether it should be. Each of five variants encodes both sequences with shared weights, preserves the time axis, and forms the same per-timestep comparison tensor for the same downstream CNN and classifier. The control, *Raw*, uses an identity encoder and reproduces the deployed model exactly at 133,058 parameters. Four *Raw* + *X* variants concatenate the encoded comparison onto the raw one, so each contains the deployed model’s evidence and can only justify itself by improving on it. *Hybrid only* removes the raw comparison, isolating what the encoder contributes alone.

The result is unambiguous and it is negative (Table II, Fig. 4, right). The deployed model is the best of the five, at 94.25% $\pm$ 3.09% ROC-AUC. Because all variants share folds and seeds the comparison is pairwise, removing fold difficulty and initialisation luck. The paired deficits range from -2.53 points for *Raw* + *CNN* to -5.93 for *Raw* + *BiLSTM*; all four are significant at $p < 10^{-3}$ (two-sided paired *t*-test, $n = 39$), no variant wins more than 9 of its 39 paired trials, and every 95% CI lies wholly below zero. Adding capacity does not merely fail to help: the largest variant, at 981,000 parameters, is 4.22 points worse than the 133,058-parameter control.

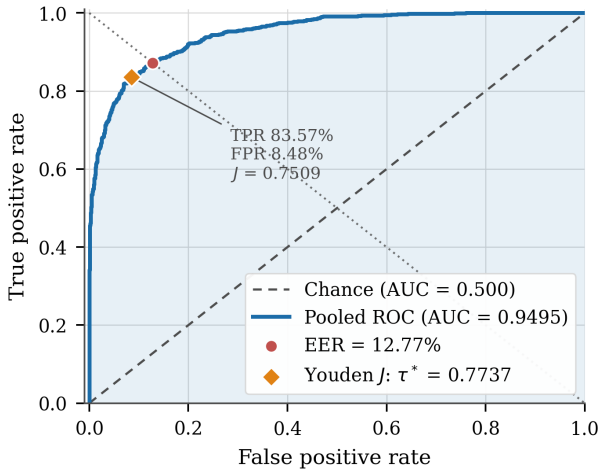


Fig. 4. Left: pooled ROC over all 2,240 leave-one-subject-out verification pairs, with the equal-error point and the Youden-optimal point $\tau^* = 0.7737$ marked. Right: paired change in ROC-AUC against the deployed model for each ablation variant, with 95% confidence intervals; every interval lies wholly below zero.

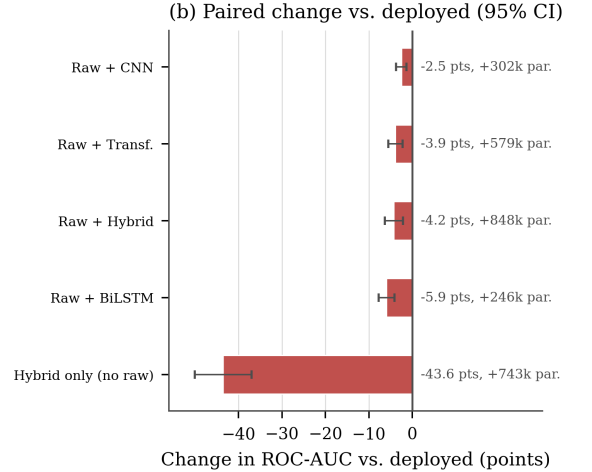


TABLE II
LOOCV ABLATION OVER 13 FOLDS AND 3 SEEDS ($n = 39$ PAIRED OBSERVATIONS PER VARIANT). Δ IS THE PAIRED CHANGE IN ROC-AUC AGAINST THE DEPLOYED MODEL

Variant	Params	ROC-AUC (%)	Acc (%)	Δ AUC
Raw (deployed)	133,058	94.25 $\pm$ 3.09	86.04	—
Raw + CNN	434,690	91.72 $\pm$ 4.13	82.66	-2.53*
Raw + Transformer	712,002	90.37 $\pm$ 5.13	80.32	-3.88*
Raw + Hybrid	980,994	90.03 $\pm$ 6.11	81.26	-4.22*
Raw + BiLSTM	379,074	88.32 $\pm$ 6.08	74.84	-5.93*
Hybrid only	876,162	50.70 $\pm$ 20.81	49.37	-43.55*

*Significant at $p < 10^{-3}$, two-sided paired t -test over the 39 matched (fold, seed) pairs. “Hybrid only” omits the raw comparison entirely.

Hybrid only settles which half of the architecture carries the signal. Stripped of the raw per-timestep comparison it reaches 50.70% $\pm$ 20.81% ROC-AUC—indistinguishable from chance, with a standard deviation four times any other variant’s, indicating it often fails to learn at all. The discriminative information is in the direct per-dimension comparison, where dimension j of each sequence is the same physical quantity and subtraction is meaningful; passing both through a learned encoder first destroys that correspondence and forces the classifier to recover it from what begins as a random projection. The ablation thus justifies the deployed architecture: the inert branch of Section III-E2 is not a missed opportunity, and a richer temporal encoder is here a reliable regression, not a free improvement.

D. Face-Swap Video Validation

Three face-swapped clips were generated as described in Section IV, pairing enrolled subjects so that both body and face sources have enrolled signatures. Each clip was scored against the claimed (face) identity at $\tau^* = 0.7737$ and re-scored against every enrolled identity to find the closest match. All three were correctly rejected (Table III): the claimed identity scores below 0.006—four to six orders of magnitude below τ^* —while the true body source scores above 0.9998, exactly as Section III-A predicts. Pose extraction succeeded over the full length of all three clips (135, 149 and 327 valid

TABLE III
FACE-SWAP VIDEO VALIDATION ($n = 3$; EXISTENCE CHECK, NOT A STATISTICALLY POWERED RESULT)

Clip (body_face)	Claimed	Sim.	Matched	Sim.
Subj.6_Subj.3	Subj. 3	0.0057	Subj. 6	0.9999
Subj.7_Subj.11	Subj. 11	0.0003	Subj. 7	0.9999
Subj.8_Subj.5	Subj. 5	<0.0001	Subj. 8	0.99999*

*Nearest impostor score for this clip (Subject 10) was 0.99986, the closest margin observed; see discussion in text.

frames), so this is not an artefact of partial pose failure. One caveat: for Subject 8_body_Subject 5_face the correct match (Subject 8, 0.99999) is nearly tied by an impostor score for Subject 10 (0.99986), the two nearest scores in the validation; the clip is still handled correctly, but the embedding space is not uniformly well separated across subject pairs.

Three clips from a single pipeline under one masking configuration is an existence check—not a statistically powered validation, silent on other face-swap models, enhancement post-processing or heavier recompression, and too small to establish how often near-collisions occur.

VI. EXPLAINABILITY

To determine what the verification network uses, we compute gradient-times-input attribution on the 78-dimensional descriptor. For a sample scored towards class c we backpropagate the corresponding logit to the input and take $A_{t,j} = |\hat{f}_{t,j} \cdot \partial \ell_c / \partial \hat{f}_{t,j}|$, aggregating over timesteps and over each joint’s three coordinates, then normalising to $[0, 1]$, averaged over 26 samples (two per identity) [29]. Because the gradient path runs through the verification head, this describes the function that actually produces the verdict.

A. Joint-Level Attribution

The left shoulder receives the highest attribution (1.000; Fig. 5), followed by the right heel (0.940), left foot index (0.931) and left knee (0.897); both hips rank lowest (0.665 left, 0.566 right).

The structure is biomechanically coherent: shoulders, carrying the trunk counter-motion that balances angular momentum during swing, and the distal foot segments, registering heel strike and toe-off, dominate [13], [14], while the hips rank lowest as expected, since the pelvis centre is the origin of the coordinate normalisation. We do not over-interpret the left-right asymmetry: with 13 subjects and 26 samples it is as consistent with cohort-specific or viewpoint bias as with anything learned about laterality.

B. Feature-Group Attribution

Attribution decomposes by feature block as follows (Fig. 6). Coordinates take 47.68%, velocities 37.40% and joint angles 14.92%, against dimensional shares of 46.15%, 46.15% and 7.69%: per-dimension ratios of 1.03, 0.81 and 1.94. The six angle features are thus roughly twice as influential as the average coordinate and nearly two and a half times the average velocity. The six scale-free, rotation-invariant angles are the most information-dense part of the descriptor, suggesting a cheaper angle-weighted feature set may retain most of the discriminative power, relevant to embedded deployment.

VII. DISCUSSION

A. Comparative Positioning

We deliberately report no head-to-head table: the figures quoted in Section efsec:related each use a different corpus, protocol and task, and our 94.95% is a leave-one-subject-out verification result on a 13-subject corpus collected by the authors. Tabulating them would invite a ranking they cannot support; no claim of superiority is made. Gait evidence has begun to appear in forensic practice: a 2025 trade-press account reports a commercial analysis accepted in a European murder case where the face occupied roughly 10imes5 pixels [30], indicating institutional interest, not a validated standard (the sub-0.1% EER is a vendor figure).

B. What This Approach Buys

Four properties follow from the modality rather than the model. *Independence from synthesis quality*: the detector never observes the manipulated region, so better face-swap fidelity does not degrade it, breaking the arms-race coupling that constrains artefact-based methods [1], [4]. *Uncorrelated failure modes*: the evidence is disjoint from that used by facial detectors, so the two families fail on different inputs—making fusion natural not because gait is stronger but because it is independent. *Privacy*: video is reduced to 12 skeletal coordinates per frame, retaining no facial imagery (though gait is itself a biometric). *Degradation profile*: gait stays recoverable at distances where facial analysis is not viable [30].

C. Limitations and Threats to Validity

We list these in descending order of how much they should temper the results. *The face-swap validation is small*. 3/3 clips were rejected with true-source similarity above 0.9998, but $n = 3$ from one generator under one masking configuration is an existence check that establishes no detection rate or

confidence interval; several dozen clips across at least two generators is the most important gap. *Thirteen subjects is a small cohort*. Augmentation multiplies sequences, not gaits, so the effective sample is 13; LOOCV at this size carries wide confidence intervals [26], reflected in the ± 3.08 AUC and ± 6.82 recall standard deviations, and CASIA-B-scale validation is needed before these are treated as population estimates. The cohort is also demographically narrow (students aged roughly 19–25; age-related and pathological gait are unrepresented). *The ablation is specific to this scale*. Twelve training subjects is a regime that penalises capacity, so the finding is not a general claim that temporal encoders cannot help gait verification, and whether the ordering reverses on a CASIA-B-scale cohort is untested. *No standard-benchmark comparison*. We report no FaceForensics++, Celeb-DF or DFDC figure: those corpora are face-centred and lack the full-body walking this method requires. *Conditions are controlled*: indoor, cooperative, unobstructed full-body framing, two viewpoints; occlusion, crowding, oblique angles, carried loads and varied footwear are unevaluated. *The threat model has a boundary*. A generator that re-synthesises, retimes or warps whole-body motion violates its assumption; such systems exist and are the subject of emerging benchmarks [18], [19], and current generative animation appears not to preserve gait identity cues faithfully [21], which suggests—but does not show—that gait may remain discriminative against full-body synthesis. *Scope of the verdict*. The system adjudicates a claimed identity, requires enrolment and cannot flag a synthetic person claiming none; it is a verification component, not a general-purpose detector, and its scores are not calibrated probabilities.

D. Future Work

Priorities: quantify detection on many more forgeries across multiple face-swap models, extend the cohort, and explore graph convolutional modelling [10], per-subject threshold calibration (Section efsec:perfold) and an angle-weighted descriptor for embedded screening.

VIII. CONCLUSION

We have presented a face-swap deepfake detector that verifies identity from skeletal gait rather than facial artefacts, exploiting the fact that face-region-only manipulation leaves the source body’s locomotion intact. A difference-based temporal convolutional network over 78-dimensional descriptors from 12 MediaPipe landmarks attains a pooled ROC-AUC of 94.95%, accuracy of $87.04\% \pm 3.65\%$ and an equal error rate of $12.27\% \pm 3.80\%$ across 13 leave-one-subject-out folds and 2,240 verification pairs; the minimal network beats every encoder configuration tested, and stripping the raw comparison reduces it to chance. The face-swap validation correctly rejected all three clips (true-source similarity above 0.9998) but remains an existence check on a single generator, thirteen subjects is a small basis for generalisation, and the ablation ordering is established at this cohort size only. Looking at

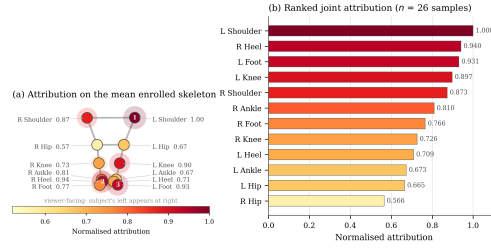


Fig. 5. Joint-level attribution (a) on the mean enrolled skeleton and (b) ranked ($n = 26$ samples).

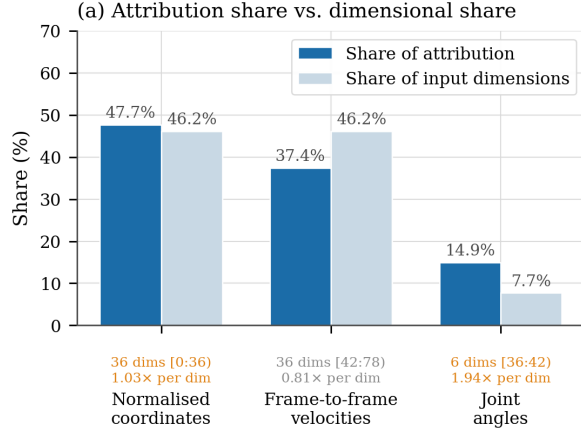


Fig. 6. Share of attribution against share of input dimensions for each feature block. Joint angles occupy 7.69% of the input but draw 14.92% of attribution, a per-dimension ratio of 1.94.

what the adversary declines to model changes the terms of the problem, and body motion is, for now, exactly that.

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REFERENCES

- [1] J. Cao, C. Ma, T. Yao, S. Chen, S. Ding, and X. Yang, “End-to-end reconstruction-classification learning for face forgery detection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 4113–4122.
- [2] H. Zhao, W. Zhou, D. Chen, T. Wei, W. Zhang, and N. Yu, “Multi-attentional deepfake detection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 2185–2194.
- [3] Y. Zheng, J. Bao, D. Chen, M. Zeng, and F. Wen, “Exploring temporal coherence for more general video face forgery detection,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 15 044–15 054.
- [4] Z. Wang, J. Bao, W. Zhou, W. Wang, and H. Li, “AltFreezing for more general video face forgery detection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 4129–4138.
- [5] A. A. Hasanaath, H. Lugman, R. Katib, and S. Anwar, “FSBI: Deepfakes detection with frequency enhanced self-blended images,” *arXiv preprint arXiv:2406.08625*, 2024.
- [6] N. A. Chandra, R. Murtfeldt, L. Qiu *et al.*, “Deepfake-eval-2024: A multi-modal in-the-wild benchmark of deepfakes circulated in 2024,” *arXiv preprint arXiv:2503.02857*, 2025.
- [7] FaceFusion Contributors, “FaceFusion: Industry-leading face manipulation platform,” <https://github.com/facefusion/facefusion>, 2024, accessed 2026-08-27.
- [8] InsightFace Contributors, “InsightFace: 2d and 3d face analysis project (InSwapper face-swap model),” <https://github.com/deepinsight/insightface>, 2023, accessed 2026-08-27.
- [9] R. Chen, X. Chen, B. Ni, and Y. Ge, “SimSwap: An efficient framework for high fidelity face swapping,” in *Proc. ACM Int. Conf. Multimedia (ACM MM)*, 2020, pp. 2003–2011.
- [10] T. Teepe, A. Khan, J. Gilg, F. Herzog, S. Hörmann, and G. Rigoll, “GaitGraph: Graph convolutional network for skeleton-based gait recognition,” in *Proc. IEEE Int. Conf. Image Processing (ICIP)*, 2021, pp. 2314–2318.
- [11] A. Cătruță, A. Cosma, and E. Radoi, “GaitPT: Skeletons are all you need for gait recognition,” in *Proc. IEEE Int. Conf. Automatic Face and Gesture Recognition (FG)*, 2024, arXiv:2308.10623.
- [12] A. Cosma and E. Radoi, “Learning gait representations with noisy multi-task learning,” *Sensors*, vol. 22, no. 18, p. 6803, 2022.
- [13] J. Perry and J. M. Burnfield, *Gait Analysis: Normal and Pathological Function*, 2nd ed. SLACK Incorporated, 2010.
- [14] D. A. Winter, *Biomechanics and Motor Control of Human Movement*, 4th ed. John Wiley & Sons, 2009.
- [15] ISO/IEC, “ISO/IEC 19795-1:2021 — information technology — biometric performance testing and reporting — part 1: Principles and framework,” International Organization for Standardization, 2021.
- [16] S. Agarwal, H. Farid, Y. Gu, M. He, K. Nagano, and H. Li, “Protecting world leaders against deep fakes,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW), Media Forensics*, 2019, pp. 38–45.
- [17] M. Boháček and H. Farid, “Protecting world leaders against deep fakes using facial, gestural, and vocal mannerisms,” *Proceedings of the National Academy of Sciences*, vol. 119, no. 48, p. e2216035119, 2022.
- [18] —, “Human action CLIPs: Detecting AI-generated human motion,” *arXiv preprint arXiv:2412.00526*, 2024.
- [19] Z. Yang, H. Zhang, J. Zhou, C. Duan, and X. Cao, “TT-DF: A large-scale diffusion-based dataset and benchmark for human body forgery detection,” *arXiv preprint arXiv:2505.08437*, 2025.
- [20] M. Chapariniya, T. Vukovic, S. Ebling, and V. Schmid, “Two-stream spatial-temporal transformer framework for person identification via natural conversational keypoints,” *arXiv preprint arXiv:2502.20803*, 2025.
- [21] I. DeAndres-Tame, M. Ye, R. Tolosana, R. Vera-Rodriguez, and S. Yu, “Is visual realism enough? evaluating gait biometric fidelity in generative AI human animation,” *arXiv preprint arXiv:2512.19275*, 2025.
- [22] A. Penwala, A. Bharti, A. A. R. Kodeeswaran, V. Sharma, and D. R. Johnson, “Deepfake detection using gait analysis (GaitDeepfake-13),” 2026, IEEE DataPort.
- [23] Google, “MediaPipe pose landmarker,” <https://ai.google.dev/edge/mediapipe/solutions/pose/landmarker>, 2023, accessed 2026-08-27.
- [24] J. Stenum, C. Rossi, and R. T. Roemmich, “Two-dimensional video-based analysis of human gait using pose estimation,” *PLOS Computational Biology*, vol. 17, no. 4, p. e1008935, 2021.
- [25] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.
- [26] G. Varoquaux, “Cross-validation failure: Small sample sizes lead to large error bars,” *NeuroImage*, vol. 180, pp. 68–77, 2018.
- [27] T. Fawcett, “An introduction to ROC analysis,” *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.
- [28] W. J. Youden, “Index for rating diagnostic tests,” *Cancer*, vol. 3, no. 1, pp. 32–35, 1950.
- [29] M. Sundararajan, A. Taly, and Q. Yan, “Axiomatic attribution for deep networks,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2017, pp. 3319–3328.
- [30] C. Burt, “Gait, body structure biometrics accepted as court evidence in EU murder case,” Biometric Update, Sep. 2025, reporting on Cursor Insight; accessed 2026-08-27.